

Task & Order Tracking System

Implementasi REST API & n8n untuk Manajemen Pesanan Percetakan

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 <https://github.com/AdityaaWardanaa/UAS-CommProto-OrderTracking>

 <https://youtu.be/0NUGug2JsgE>

PROBLEM & USE CASE

Masalah Bisnis

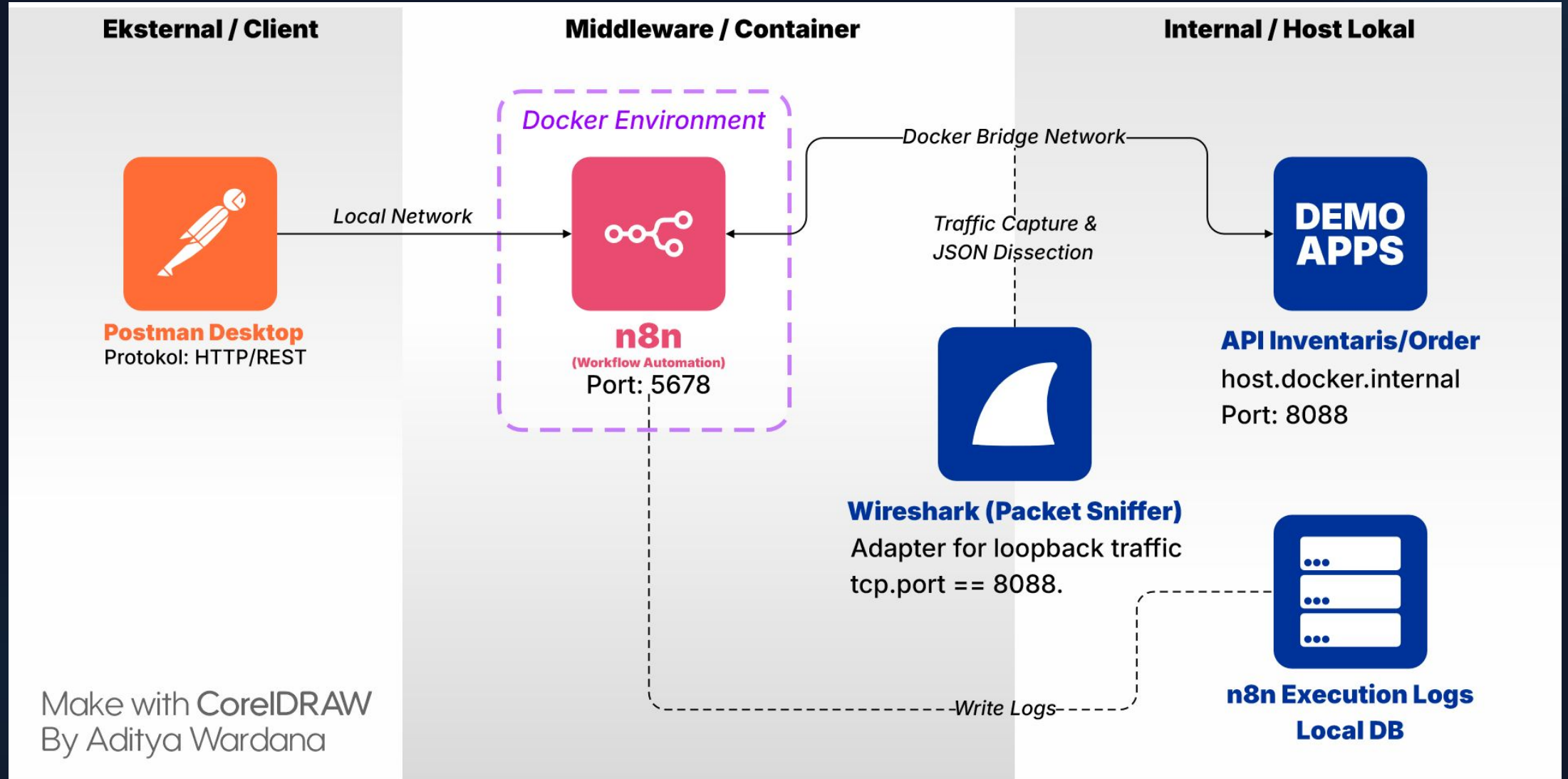
Proses manual dalam pelacakan pesanan percetakan (banner, stiker, jilid) menyebabkan inefisiensi komunikasi antara admin dan bagian produksi. Status pesanan sering kali tidak transparan bagi pelanggan.

Use Case 4: Task/Order Tracking

Sistem otomatisasi untuk mencatat, memperbarui, dan melacak status pesanan secara real-time.



ARCHITECTURE CANVAS



ENDPOINT & ACTION DESIGN

Method	Endpoint	Action / Description	Status Code
POST	/api/orders	Membuat pesanan cetak baru (Payload JSON)	201 Created
GET	/api/orders	Menarik seluruh riwayat pesanan (Listing)	200 OK
GET	/api/orders/{id}	Melacak status pesanan spesifik (Tracking)	404 if invalid

Inisiator: **Postman**
Desktop

Gateway: **n8n Webhook**

TESTING EVIDENCE



Success: 201

JSON Payload pesanan banner berhasil diproses dan disimpan ke database lokal melalui n8n.



Failure: 400

API menolak request karena 'Invalid Fields' (Atribut JSON tidak sesuai dengan skema allowed).



Failure: 404

Pencarian order ID fiktif (99999) mengembalikan respons 'Order Not Found'.

OBSERVABILITY & RELIABILITY



n8n Execution Logs: Merekam riwayat traffic secara otomatis (timestamp, payload, response).



Wireshark Sniffing: Validasi integritas data pada layer network (TCP Port 8088).



Reliability: API tetap stabil dan mampu memberikan pesan error informatif saat input tidak valid.

The screenshot displays a network simulation interface with a menu bar (Berkas, Sunting, Lihat, Go, Tangkap, Analisis, Statistik, Teleponi, Nirkabel, Perangkat, Bantuan) and a toolbar. The main window shows a list of network packets with columns for No., Time, Source, Destination, Protocol, Length, and Info. The selected packet (No. 432) is an HTTP POST request to /api/orders. Below the packet list, the 'Packet Details' pane shows the structure of the request, including headers like 'Host: localhost:8088' and 'Content-Type: application/json', and a body with order details. The 'Packet Bytes' pane shows the raw hex and ASCII data of the packet. The bottom status bar indicates the simulation is running on 'task order uas simulation communication protocols.pcapng'.

CONCLUSION & FUTURE

Kesimpulan

Integrasi n8n dan REST API berhasil mengotomatisasi alur pelacakan pesanan dengan visibilitas data yang tinggi (Observability).

Limitation & Improvement

-  **Limit:** Belum ada layer autentikasi (API Key/JWT).
-  **Improve:** Implementasi Notifikasi WhatsApp via n8n saat status pesanan berubah.
-  **Improve:** Migrasi database ke PostgreSQL container.